

Bayesian Analysis

and

Bayesian Learning

Learning Objectives

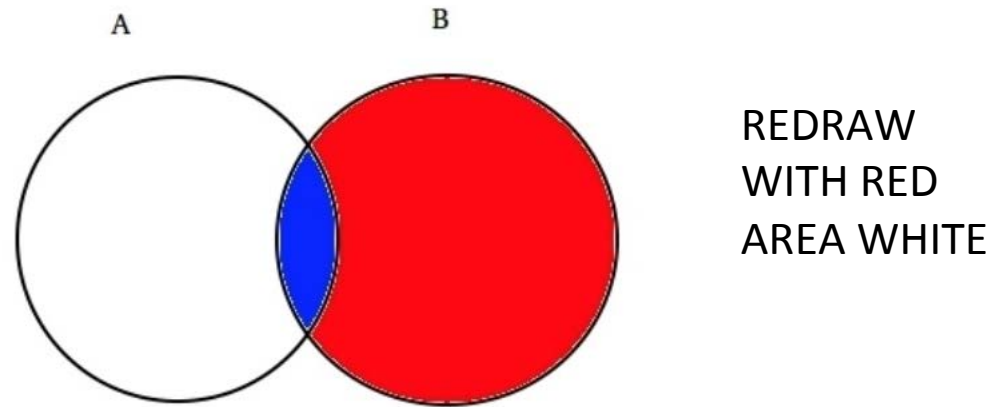
1. Be able to apply Bayes theorem to predict the probabilities of various potential outcomes using increasingly specific empirical data.
2. Be able to describe and implement the following Bayesian machine learning models:
 - A. Maximum a posteriori (MAP) hypothesis classifier
 - B. Maximum likelihood (ML) hypothesis classifier
 - C. Bayes optimal classifier
 - D. Bayes optimal classifier using Gibb's algorithm
 - E. Naïve Bayes classifier

Bayes Theorem

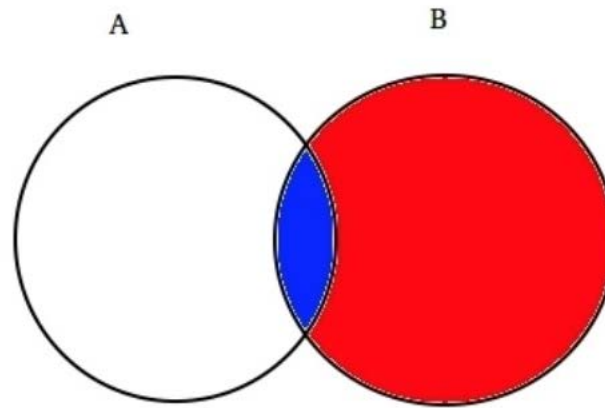
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)},$$

where A and B are events and $P(B) \neq 0$.

- $P(A|B)$ is a **conditional probability**,
the likelihood of event A occurring given that B is true.
- $P(B|A)$ is also a conditional probability,
the likelihood of event B occurring given that A is true.
- $P(A)$ and $P(B)$ are the probabilities of observing A and B
independently of each other, which are known as the
marginal probabilities of A and B .



Consider two events, A and B. In this diagram the circle on left represents the probability that A occurs and the circle on the right the probability that event B occurs. The shaded area is the probability that both A and B occur.



Lets assume that event B has occurred; we'll shade that circle. The probability that event A has occurred given that event B has occurred is the the probability of the intertecting area divided by the sum of the probabilities represented by the two shaded areas,

$$P(A|B) = \frac{P(A \cap B)}{P(A)}.$$

Medical diagnosis example for a rare cancer

General Population

Cancer	Not Cancer
$P(\text{Cancer})=0.001$	$P(\neg \text{Cancer})=0.999$

Measures of the accuracy of a test for this cancer

RISE

Confusion Matrix

	Cancer	Not Cancer
Pos Test	True Positive	False Positive
NegTest	False Negative	True Negative

Measures of the accuracy of a test for this cancer

Confusion Matrix

	Cancer	Not Cancer
Pos Test		
NegTest		

STOP

A patients tests positive

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If a patient's test result is $\oplus$, what is probability that the patient has the disease?

$$\begin{aligned}P(\text{cancer} \mid \oplus) &= P(\oplus \mid \text{cancer}) P(\text{cancer}) \\&= (0.98)(0.001) = 0.00098\end{aligned}$$

$$\begin{aligned}P(\neg \text{cancer} \mid \oplus) &= P(\oplus \mid \neg \text{cancer}) P(\neg \text{cancer}) \\&= (0.03)(0.999) = 0.02997\end{aligned}$$

Medical diagnosis example ...

$$P(\text{cancer} \mid \oplus) = 0.00098$$

$$P(\neg\text{cancer} \mid \oplus) = 0.02997$$

Since for individual patient either had cancer or not,

$$P_{\oplus}(\text{cancer}) + P_{\oplus}(\neg\text{cancer}) = 1.$$

Thus,

$$P_{\oplus}(\text{cancer}) = \frac{0.00098}{0.00098 + 0.02997} = 0.0317$$

$$P_{\oplus}(\neg\text{cancer}) = \frac{0.02997}{0.00098 + 0.02997} = 0.9683$$

Confusion Matrix

		Cancer	Not Cancer
Pos Test		True Positive	False Positive
NegTest		False Negative	True Negative

Confusion Matrix

		Cancer	Not Cancer
		$P(\text{cancer})$	$P(\text{not cancer})$
Pos Test	$P(\text{pos test})$	$P(\text{pos test} \ \& \ \text{cancer})$	$P(\text{pos test} \ \& \ \text{not cancer})$
NegTest	$P(\text{neg test})$	$P(\text{neg test} \ \& \ \text{cancer})$	$P(\text{neg test} \ \& \ \text{not cancer})$

Confusion Matrix

		Cancer	Not Cancer
		$P(\text{cancer})$	$P(\text{not cancer})$
Pos Test	$P(\text{pos test})$	$P(\text{pos test} \text{cancer}) * P(\text{cancer})$	$P(\text{pos test} \text{not cancer}) * P(\text{not cancer})$
NegTest	$P(\text{neg test})$	$P(\text{neg test} \text{cancer}) * P(\text{cancer})$	$P(\text{neg test} \text{not cancer}) * P(\text{not cancer})$

Why is $P_+(cancer)$ only 0.0317?

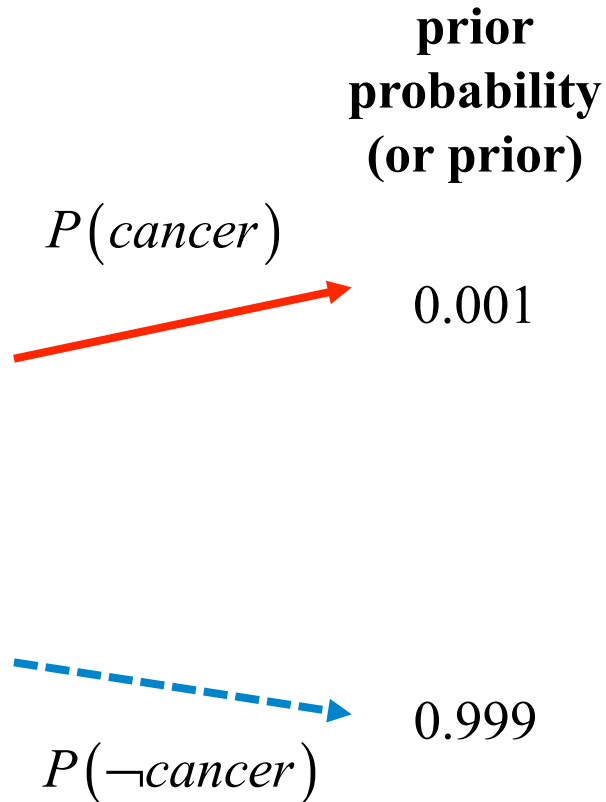
Why is $P_+(cancer)$ only 0.0317?

		Cancer	Not Cancer
	100,000	100	99,900
Pos Test	3,095	98	2,997
NegTest	96,905	2	96,903

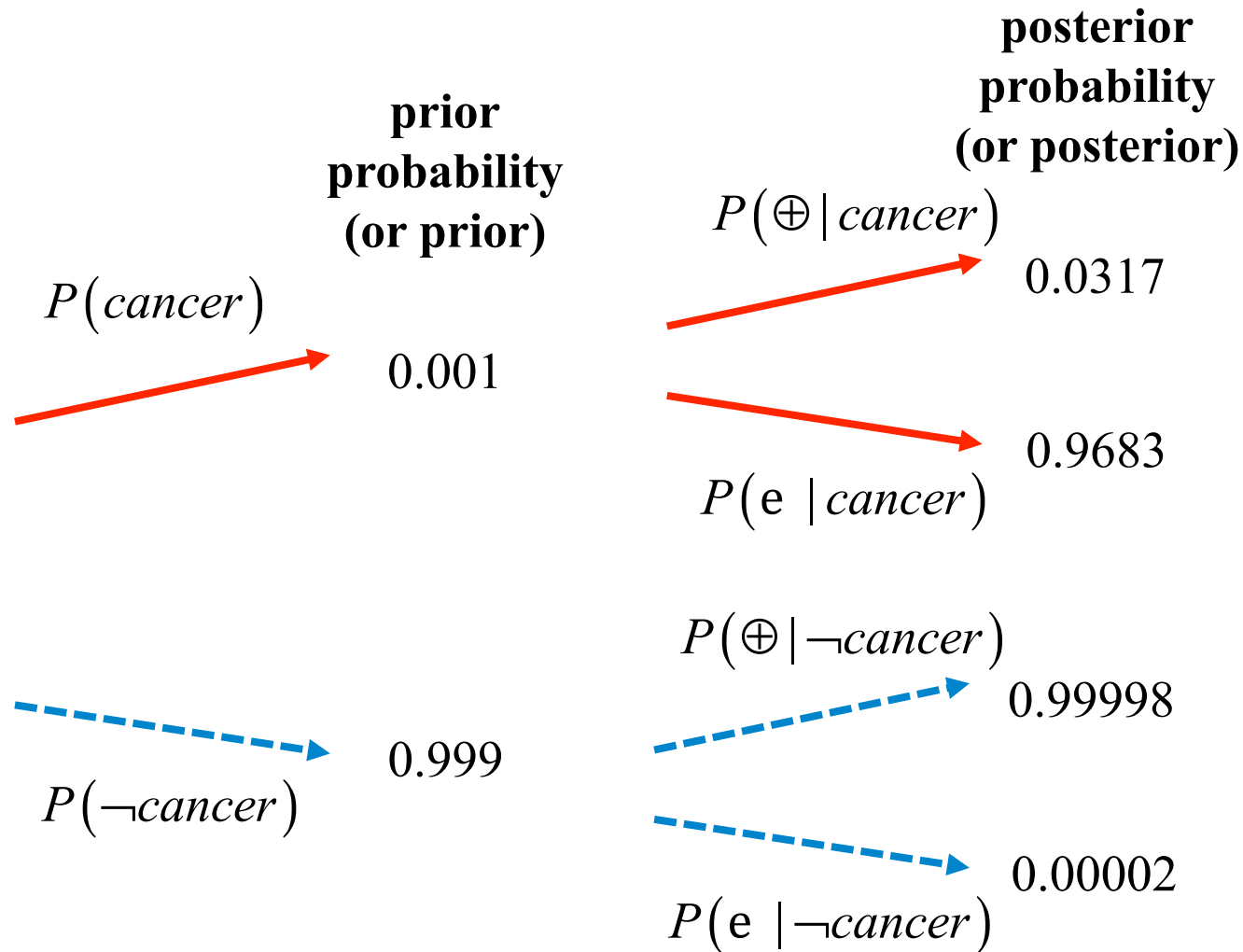
Why is $P_+(cancer)$ only 0.0317?

			Cancer	Not Cancer
			0.1%	99.9%
		100,000	100	99,900
Pos Test	3.095%	3,095	98	2,997
NegTest	96.905%	96,905	2	96,903

Prior and posterior probabilities

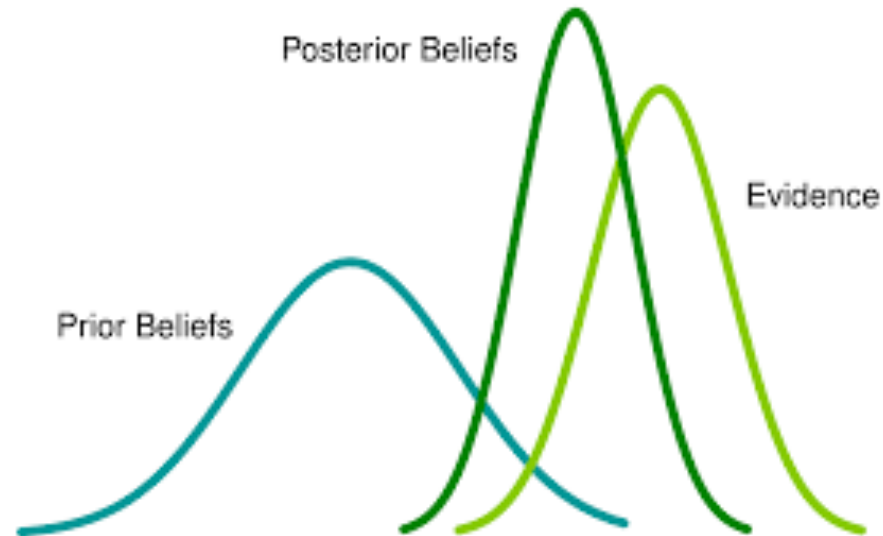


Prior and posterior probabilities



STOP

Prior and posterior probabilities



$$P(e \mid \neg cancer)$$

Thomas Bayes

RISE



1702-1761

Thomas Bayes

- English statistician, philosopher, and Presbyterian minister
- Educated at the University of Edinburgh in logic and theology
- Is known to have published two works:
 - *Divine Benevolence, or an Attempt to Prove That the Principal End of the Divine Providence and Government is the Happiness of His Creatures* (1731)
 - *An Introduction to the Doctrine of Fluxions, and a Defence of the Mathematicians Against the Objections of the Author of The Analyst* (published anonymously in 1736), in which he defended the logical foundation of Isaac Newton's calculus ("fluxions") against the criticism of George Berkeley, author of *The Analyst*
- Bayes' work on the "Problem of inverse probability" was presented in 1763 to the Royal Society by Richard Price following Bayes' death two years earlier.

Back to Bayes Theorem

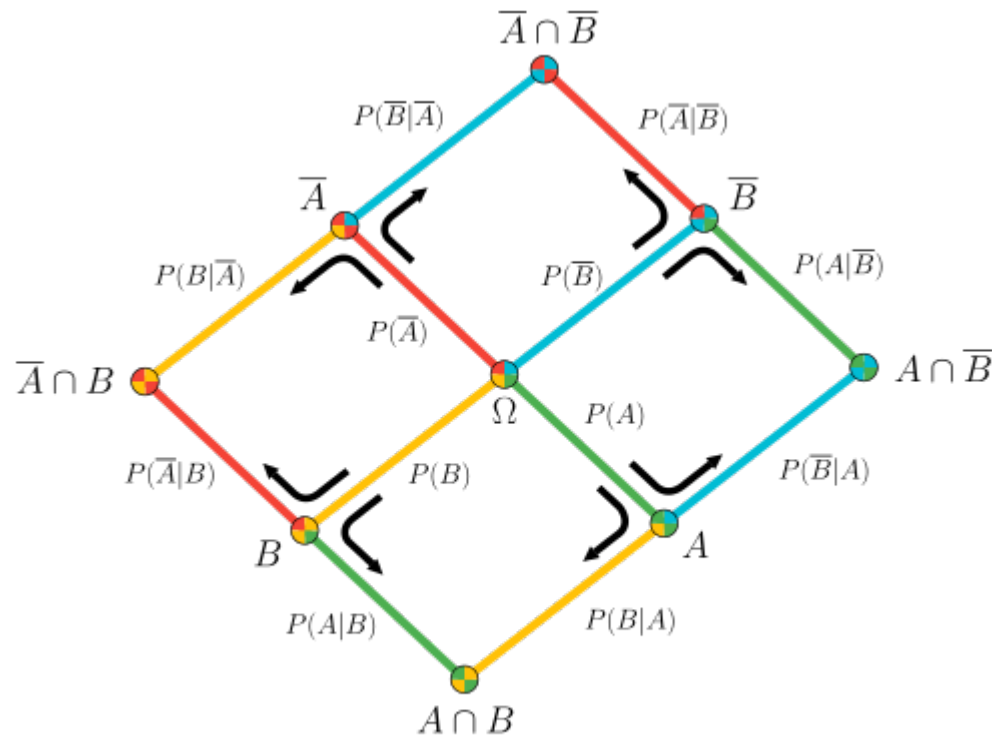
$$\text{Since } P(A|B) = \frac{P(B|A)P(A)}{P(B)},$$

$$P(A|B)P(B) = P(B|A)P(A).$$

$$P(A|B)P(B) = P(A \cap B) = P(B \cap A) = P(B|A)P(A).$$

A nice graphical representation

RISE



$$P(A|B) \cdot P(B) = P(A \cap B) = P(B|A) \cdot P(A)$$

By Qniemiec - https://commons.wikimedia.org/wiki/File:Bayes%27_Theorem_2D.png, CC BY-SA 3.0,
<https://commons.wikimedia.org/w/index.php?curid=44778639>

from Wikipedia

A twins example

A physicist is pregnant and her doctor tells her that she's having twins. She asks what's the probability that they will be **identical** rather than **fraternal**. There are two mutually exclusive hypotheses:

- g The twins are identical,

- g The twins are not identical (and, therefore, fraternal).

Her doctor tells her that one-third of twin births are identical and two-thirds are fraternal.

Thus,

$$P(I) = \frac{1}{3}$$

$$P(F) = \frac{2}{3}.$$

The twins example with more information

The physicist has a sonogram which indicates that her twins will be of the same sex (boys, to be specific).

Again, there are two mutually exclusive hypotheses:

- g The twins are of the **same** sex,

- g The twins are of **different** sexes.

Prior to receiving the results of the sonogram,

$$P(S) \approx \frac{1}{2},$$

$$P(D) \approx \frac{1}{2}.$$

But, given the results of the sonogram,

$$P(S) = 1,$$

$$P(D) = 0.$$

Applying Bayes theorem

$$P(I | S) = P(S | I)P(I) = 1 \cdot \frac{1}{3} = \frac{1}{3}$$

$$P(F | S) = P(S | F)P(F) = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3},$$

Normalizing we get

$$P_{Norm}(I | S) = \frac{P(I | S)}{P(I | S) + P(F | S)} = \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{3}} = \frac{1}{2},$$

$$P_{Norm}(F | S) = \frac{P(F | S)}{P(I | S) + P(F | S)} = \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{3}} = \frac{1}{2}.$$

Prior and posterior probabilities

The prior probabilities are

$$P_{prior}(I) = \frac{1}{3}$$

$$P_{prior}(F) = \frac{2}{3},$$

while the posterior probabilities are

$$P_{posterior}(I) = \frac{1}{2},$$

$$P_{posterior}(F) = \frac{1}{2}.$$

Another approach to the twins example

		Sonogram Results		
		Same sex	Different	
Twins are	Identical	a	b	
	Fraternal	c	d	

- $a+b+c+d=1$ since these are probabilities

Another approach ...

		Sonogram Results		
		Same sex	Different	
Twins are	Identical	a	b=0	
	Fraternal	c	d	

- $b=0$ since identical twins cannot be of different sexes

Another approach ...

		Sonogram Results		
		Same sex	Different	
Twins are	Identical	a	0	
	Fraternal	c	d	

- $c=d$ since the sexes of fraternal twins is random

Another approach ...

		Sonogram Results		Doctor's Prior
		Same sex	Different	
Twins are	Identical	a	0	1/3
	Fraternal	c	d	2/3

- $P(\text{Identical}) = 1/3$
- $P(\text{Fraternal}) = 2/3$

based on a large database upon which the Doctor makes his/her statement.

Another approach ...

		Sonogram Results		Doctor's Prior
		Same sex	Different	
Twins are	Identical	a	b=0	1/3
	Fraternal	c	d	2/3

- $a+b = 1/3$
- $b = 0$
- $c + d = 2/3$
- $c = d = 1/3$

Another approach ...

		Sonogram Results		Doctor's Prior
		Same sex	Different	
Twins are	Identical	1/3	0	1/3
	Fraternal	1/3	1/3	2/3

Another approach ...

		Sonogram Results		Doctor's Prior
		Same sex	Different	
Twins are	Identical	1/3	0	1/3
	Fraternal	1/3	1/3	2/3
		Physicist		

Another approach ...

		Sonogram Results		Doctor's Prior	Physicist's Posterior
		Same sex	Different sexes		
Twins are	Identical	1/3	0	1/3	1/2
	Fraternal	1/3	1/3	2/3	1/2

Bayes Theorem in machine learning

We consider a set $H = \{h_1, h_2, \dots\}$ of candidate hypotheses from which we seek to find the "best" one.

A hypothesis is just a set of rules that we use to predict an output for a given input.

There is no single definition of "best"; different choices of best lead to different algorithms.

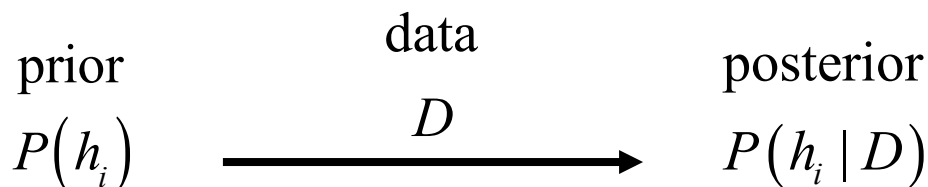
Bayes theorem can be used to choose among candidate hypotheses and when it is used for this purpose, the resulting algorithms are referred to as Bayesian learning. Different algorithms result not only from different choices of "best," but also from what simplifying assumptions, if any, are made.

Bayes Theorem in machine learning

$P(h_i)$ is the prior (probability) that we believe hypothesis h_i to be correct

$P(h_i | D)$ is the posterior (probability) that we believe hypothesis h_i to be correct having accounted for the data D .

$\frac{P(D | h_i)}{P(D)}$ is the support that D provides for h_i .



Marginal probabilities

$P(h)$ is probability that hypothesis h holds.

- g Often referred to as the **prior probability of h** .

- g Assumes no knowledge of the training data D

$P(D)$ is the probability that the training data D is observed.

- g Often referred to as the **prior probability of D** .

- g Assumes no knowledge of which hypotheses h is correct.

STOP

Bayesian Machine Learning Models

Classifier	Summary Description
Maximum a Posteriori Hypothesis	Use most probable hypothesis
Maximum Likelihood	
Bayes Optimal	
Bayes Optimal using Gibbs Algorithm	
Bayes Naïve	

Maximum a posteriori (MAP) learning

We can consider a set H of candidate hypotheses from which we seek to find the most probable hypotheses $h \in H$. Any such maximally probable hypothesis is referred to as a *maximum a posteriori* (MAP) hypothesis and is denoted by h_{MAP} .

We use Bayes Theorem to calculate the posterior probability of each candidate hypothesis. Thus,

$$\begin{aligned} h_{MAP} &\equiv \operatorname{argmax}_{h \in H} P(h | D) \\ &= \operatorname{argmax}_{h \in H} \frac{P(D | h) P(h)}{P(D)} \\ &= \operatorname{argmax}_{h \in H} P(D | h) P(h) \end{aligned}$$

Bayesian Machine Learning Models

Classifier	Summary Description
Maximum a Posteriori Hypothesis	Use most probable hypothesis
Maximum Likelihood	Use the hypothesis that maximizes the likelihood of the data given a hypothesis
Bayes Optimal	
Bayes Optimal using Gibbs Algorithm	
Bayes Naïve	

Maximum likelihood (ML) learning

If we can assume that every hypothesis in H is equally probable *a priori*, then $P(h_i) = P(h_j)$ for all $h_i, h_j \in H$, and the expression for h_{MAP} can be simplified.

$$h_{\text{MAP}} = \underset{h \in H}{\operatorname{argmax}} P(D | h) P(h)$$

becomes

$$h_{\text{ML}} = \underset{h \in H}{\operatorname{argmax}} P(D | h)$$

since $P(h)$ is a constant that is independent of h .

$P(D | h)$ is referred to as the **likelihood** of the data D given h and any hypothesis that maximizes $P(D | h)$ is referred to as a **maximum likelihood (ML) hypotheses**, h_{ML} .

MAP learning algorithm

Consider a finite hypothesis space H that is defined over the instance space X for which the task is to learn some target concept $c : X \rightarrow \{0,1\}$.

We assume that the learner is given a set of training examples

$\{\{x_1, d_1\}, \dots, \{x_m, d_m\}\}$, where $d_i = c(x_i)$.

The following is a simple algorithm to output a maximum a posteriori hypothesis:

- 1) For each hypothesis $h \in H$, calculate the posterior probability

$$P(h | D) = \frac{P(D | h) P(h)}{P(D)}.$$

- 2) Output the hypothesis h_{MAP} with the highest posterior probability

$$h_{MAP} = \operatorname{argmax}_{h \in H} P(h | D)$$

from Mitchell

Assumptions and the choice for $P(h)$

We need to choose $P(h)$ and $P(D | h)$ to describe our prior knowledge.

For this example we choose them to be consistent with the following assumptions:

- 1) The training data is noise free, i.e., $d_i = c(x_i)$.
- 2) The target concept is contained in the hypothesis space H .
- 3) There is no a priori reason to believe that any hypothesis is more probable than any other.

Since each hypothesis is equally probable and since their probabilities are normalized (i.e., $\sum_{h \in H} P(h) = 1$), our choice of $P(h)$ must be

$$P(h) = \frac{1}{|H|} \text{ for all } h \in H, \text{ where } |H| \text{ denotes the number of elements in } H.$$

The choice for $P(D|h)$

Since we assume that the training data is noise-free, the probability of observing classification d_i given h is 1 if D is consistent with h and 0 otherwise.

Thus,

$$P(D|h) = \begin{cases} 1 & \text{if } d_i = h(x_i) \text{ for all } d_i \in D \\ 0 & \text{otherwise.} \end{cases}$$

Applying Bayes Theorem to each h

For each hypothesis $h \in H$, calculate the posterior probability

$$P(h | D) = \frac{P(D | h)P(h)}{P(D)}.$$

If h is inconsistent with the training data D , then $P(D | h) = 0$ and

$$P(h | D) = \frac{P(D | h)P(h)}{P(D)} = \frac{0 \cdot P(h)}{P(D)} = 0.$$

If h is consistent with the training data D , then $P(D | h) = 1$

An expression for $P(D)$

We still need an expression for $P(D)$.

$$\begin{aligned} P(D) &= \sum_{h_i \in H} P(D | h_i) P(h_i) \\ &= \sum_{h_i \in VS_{H,D}} 1 \cdot \frac{1}{|H|} + \sum_{h_i \notin VS_{H,D}} 0 \cdot \frac{1}{|H|} \\ &= \sum_{h_i \in VS_{H,D}} 1 \cdot \frac{1}{|H|} \\ &= \frac{|VS_{H,D}|}{|H|}. \end{aligned}$$

where $VS_{H,D}$ is the subset of hypotheses from H that are consistent with D .

An expression for $P(h|D)$

Thus, if h is consistent with the training data D , then $P(D|h) = 1$ and

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)} = \frac{1 \cdot \frac{1}{|H|}}{\frac{|VS_{H,D}|}{|H|}} = \frac{1}{|VS_{H,D}|}.$$

MAP learning algorithm

Given our assumptions,

$$P(h | D) = \begin{cases} \frac{1}{|VS_{H,D}|} & \text{if } h \text{ is consistent with } D \\ 0 & \text{otherwise,} \end{cases}$$

where $|VS_{H,D}|$ is the number of hypotheses from H that are consistent with D .

Since the posterior probability $P(h | D)$ equals $\frac{1}{|VS_{H,D}|}$ for every consistent hypothesis and 0 otherwise, every consistent hypothesis is a MAP hypothesis

Bayesian Machine Learning Models

Classifier	Summary Description
Maximum a Posteriori Hypothesis	Use most probable hypothesis
Maximum Likelihood	Use the hypothesis that maximizes the likelihood of the data given a hypothesis
Bayes Optimal	Use the most probable outcome
Bayes Optimal using Gibbs Algorithm	
Bayes Naïve	

Bayes Optimal Classifier

A maximum a posteriori (MAP) hypothesis is the (or a) most probable hypothesis given the training data.

But does applying a MAP hypothesis to a new instance necessarily provide the most probable prediction (e.g., predicted classification) for the new instance?

Perhaps not.

A binary classification example

Hypothesis	$P(h_i D)$	$h_i(x)$
h_1	0.4	1
h_2	0.3	0
h_3	0.3	0

$h_i(x)$ is the classification returned by applying hypothesis h_i to a new instance x .

A binary classification example

Hypothesis	$P(h_i D)$	$h_i(x)$
h_1	0.4	1
h_2	0.3	0
h_3	0.3	0

$h_i(x)$ is the classification returned by applying hypothesis h_i to a new instance x .

$$h_{MAP} = \operatorname{argmax}_{h_i} P(h_i | D) = h_1$$

A binary classification example

Hypothesis	$P(h_i D)$	$h_i(x)$
h_1	0.4	1
h_2	0.3	0
h_3	0.3	0

$h_i(x)$ is the classification returned by applying hypothesis h_i to a new instance x .

$$h_{MAP} = \operatorname{argmax}_{h_i} P(h_i | D) = h_1$$

$$h_{MAP}(x) = h_1(x) = 1$$

A binary classification example

Hypothesis	$P(h_i D)$	$h_i(x)$
h_1	0.4	1
h_2	0.3	0
h_3	0.3	0

$h_i(x)$ is the classification returned by applying hypothesis h_i to a new instance x .

$$P(h(x)=1 | D) = P(h_1 | D) = 0.4$$

$$P(h(x)=0 | D) = P(h_2 | D) + P(h_3 | D) = 0.3 + 0.3 = 0.6$$

Bayes Optimal Classifier

The most probable classification of a new instance is a combination of the predictions of all hypotheses, weighted by the posterior probability of each hypothesis.

If the classification of a new instance can take on any value v_j where $v_j \in V$, then the probability $P(v_j | D)$ that the correct classification for the new instance is v_j is

$$P(v_j | D) = \sum_{h_i \in H} P(v_j | h_i) P(h_i | D).$$

The optimal classification of the new instance is the value v_j for which $P(v_j | D)$ is the greatest.

Bayesian Machine Learning Models

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Bayes Optimal using Gibbs Algorithm	Choose a hypothesis at random according the probabilities of all hypotheses
Bayes Naïve	

Bayes Optimal Classifier

$$v_{OPT} = \operatorname{argmax}_{v_j \in V} P(v_j | D) = \operatorname{argmax}_{v_j \in V} \sum_{h_i \in H} P(v_j | h_i) P(h_i | D).$$

While the Bayes optimal classifier yields the best prediction that can be obtained given the training data, it requires the posterior probability of every hypothesis and, for each new instance, the prediction for every hypothesis.

A less computationally intensive (but less accurate) method employs the Gibbs algorithm .

Gibbs algorithm

The Gibbs algorithm, in a simplified form, is

- 1) Choose a hypothesis h from H at random according to the posterior probability distribution of H .
- 2) Use h to predict the classification of the next instance x .

While using the Gibbs algorithm still requires knowing the posterior probability of every hypothesis, for each new instance it uses the prediction for only a single hypothesis rather than for all of the hypotheses.

Applying the Gibbs algorithm

Hypothesis	$P(h_i D)$	$h_i(x)$
h_1	0.4	1
h_2	0.3	0
h_3	0.3	0

RISE slides 59-64

- 1) Choose a hypothesis h from H at random according to the posterior probability distribution of over H .
- 2) Use h to predict the classification of the next instance x .

Comparing the algorithms

Hypothesis	$P(h_i D)$	$h_i(x)$
h_1	0.4	1
h_2	0.3	0
h_3	0.3	0

Comparing the algorithms

Hypothesis	$P(h_i D)$	$h_i(x)$
h_1	0.4	1
h_2	0.3	0
h_3	0.3	0

$$v_{MAP}(x) = 1$$

Comparing the algorithms

Hypothesis	$P(h_i D)$	$h_i(x)$
h_1	0.4	1
h_2	0.3	0
h_3	0.3	0

$$v_{MAP}(x) = 1$$

$$v_{OPT}(x) = 0$$

Comparing the algorithms

Hypothesis	$P(h_i D)$	$h_i(x)$
h_1	0.4	1
h_2	0.3	0
h_3	0.3	0

$$v_{MAP}(x) = 1$$

$$v_{OPT}(x) = 0$$

$$v_{Gibbs}(x) = \begin{cases} 1 & \approx 0.4 \text{ of the time} \\ 0 & \approx 0.6 \text{ of the time} \end{cases}$$

J. Willard Gibbs



1839-1903

J. Willard Gibbs

- Affiliated with Yale for all of his studies, teaching, and research; his father had taught there.
- First Ph.D. in engineering granted in the U.S. (1863)
- First chair of mathematical physics in the U.S. (1871)
- Developed the field of vector analysis
- Contributed to the fields of crystallography and planetary orbits
- Laid the foundations for the field of physical chemistry

Bayesian Machine Learning Models

Classifier	Summary Description
Maximum a Posteriori Hypothesis	Use most probable hypothesis
Maximum Likelihood	Use the hypothesis that maximizes the likelihood of the data given a hypothesis
Bayes Optimal	Use the most probable outcome
Bayes Optimal using Gibbs Algorithm	Choose a hypothesis at random according the probabilities of all hypotheses
Bayes Naïve	Assume attribute values are conditionally independent of the target value.

Naive Bayes classifier

Consider the application of a Bayes optimal classifier to a problem with n attributes (i.e., features). Given a new instance with attributes $\langle a_1, a_2, \dots, a_n \rangle$,

$$\begin{aligned} v_{MAP} &= \operatorname{argmax}_{v_j \in V} P(v_j | a_1, a_2, \dots, a_n) \\ &= \operatorname{argmax}_{v_j \in V} \frac{P(a_1, a_2, \dots, a_n | v_j) P(v_j)}{P(a_1, a_2, \dots, a_n)} \\ &= \operatorname{argmax}_{v_j \in V} P(a_1, a_2, \dots, a_n | v_j) P(v_j). \end{aligned}$$

We can estimate $P(v_j)$ simply by counting the frequency with which each target value v_j occurs in the training data.

Naive Bayes classifier

Estimating $P(a_1, a_2, \dots, a_n | v_j)$, however, requires an enormous training dataset. For our sampling to be representative of the actual distribution, we need to see multiple samples for every combination of $\langle a_1, a_2, \dots, a_n \rangle$ and v_j .

The naive Bayes classifier make the simplifying assumption that the attribute values are conditionally independent given the target value. That is, given the target value of an instance, the probability of observing $\langle a_1, a_2, \dots, a_n \rangle$ is just the product of the probabilities of observing the individual a_j ,

$$P(a_1, a_2, \dots, a_n | v_j) \approx \prod_i P(a_i | v_j).$$

from Mitchell

Naive Bayes classifier

Since

$$v_{MAP} = \operatorname{argmax}_{v_j \in V} P(a_1, a_2, \dots, a_n | v_j) P(v_j) \text{ and}$$

$$P(a_1, a_2, \dots, a_n | v_j) \approx \prod_i P(a_i | v_j),$$

we obtain

$$v_{NB} = \operatorname{argmax}_{v_j \in V} P(v_j) \prod_i P(a_i | v_j).$$

Note that the number of distinct $P(a_i | v_j)$ terms that must be estimated from the training data is just the number of distinct attribute values times the number of distinct target values.

An example: classifying text

Let the instance space X consist of all possible text documents. Thus, X contains all possible strings of words and punctuation of all possible lengths.

We are given a training set where each sample is labeled with the value of a target function $f(x)$ where $f(x) \in V$ and V is a finite set. Our goal is to learn from the training examples to predict target values for subsequent text documents.

An example: classifying text

Let's look at the target function classifying examples as interesting or uninteresting to a particular group of readers using the target values of **like** and **dislike**.

We define an attribute for each word position in the document and assign the value of that attribute to be the word in that position.

Thus, a_i is attribute i and its value is the word in position i of the document.

First paragraph of *Moby Dick*

Call me Ishmael. Some years ago - never mind how long precisely - having little or no money in my purse, and nothing particular to interest me on shore, I thought I would sail about a little and see the watery part of the world. It is a way I have of driving off the spleen and regulating the circulation. ... This is my substitute for pistol and ball. With a philosophical flourish Cato throws himself upon his sword; I quietly take to the ship. There is nothing surprising in this. If they but knew it, almost all men in their degree, some time or other, cherish very nearly the same feelings towards the ocean with me.

Moby Dick or, The Whale
by Herman Melville
1851

First paragraph of *Moby Dick*

Call me Ishmael. Some years ago - never mind how long precisely - having little or no money in my purse, and nothing particular to interest me on shore, I thought I would sail about a little and see the watery part of the world. It is a way I have of driving off the spleen and regulating the circulation. ... This is my substitute for pistol and ball. With a philosophical flourish Cato throws himself upon his sword; I quietly take to the ship. There is nothing surprising in this. If they but knew it, almost all men in their degree, some time or other, cherish very nearly the same feelings towards the ocean with me.

$a_1 = \text{"call"}$

$a_2 = \text{"me"}$

...

$a_{200} = \text{"with"}$

$a_{201} = \text{"me"}$

...

$A_{211,763} = \text{"orphan"}$

Applying a naive Bayes classifier

Let us assume that we have 512 texts that our target audience has classified as *like* and 488 that it has classified as *dislike*.

Let us further assume that we have trained our model on these 1,000 texts and that the new document that we have been given to classify is *Moby Dick*.

We return to our expression for naïve Bayes classification:

$$v_{NB} = \operatorname{argmax}_{v_j \in V} P(v_j) \prod_i P(a_i | v_j).$$

Applying a naive Bayes classifier ...

$$\begin{aligned} v_{NB} &= \operatorname{argmax}_{v_j \in V} P(v_j) \prod_i P(a_i | v_j) \\ &= \operatorname{argmax}_{v_j \in \{like, dislike\}} P(v_j) P(a_1 = "call" | v_j) P(a_2 = "me" | v_j) \dots \\ &\quad P(a_{211,763} = "orphan" | v_j) \\ &= \operatorname{argmax}_{\{like, dislike\}} \left[\begin{array}{l} \left[P(like) P(a_1 = "call" | like) P(a_2 = "me" | like) \dots \right. \\ \quad \left. (a_{211,763} = "orphan" | like) \right] , \\ \left[P(dislike) P(a_1 = "call" | dislike) P(a_2 = "me" | dislike) \dots \right. \\ \quad \left. (a_{211,763} = "orphan" | dislike) \right] \end{array} \right] \end{aligned}$$

How naive is our naive Bayes classifier?

Very.

For example, the probabilities of finding specific words is not independent of the words immediately preceding them. “White” is immediately followed by “whale” more frequently than would random and “Moby” is immediately followed by “Dick” than would be random.

And articles (e.g., a, an, the) are almost always immediately followed by nouns rather than by verbs.

But that’s OK.

What about the conditional probabilities?

Let $w_k \in W$, where W is the set of distinct words in English.

Note that $|W| \approx 200,000$. Since $|V| = 2$ and $|a| \approx 200,000$,

we need to calculate the approximately $200,000 \cdot 2 \cdot 200,000 =$

8×10^{10} values of $P(a_i = w_k | v_j)$. But our training sample of

1,000 texts contains only roughly $1,000 \cdot |a| = 1,000 \cdot 200,000 =$

2×10^8 words and so is too small by many orders of magnitude

for us to calculate the $P(a_i = w_k | v_j)$ unless we make additional

simplifying assumptions.

A further simplification

A very reasonable assumption that we can make is that the probability of encountering a specific word w_k is independent of the position being considered, a_i , that is, that

$$P(a_i = w_k | v_j) = P(a_m = w_k | v_j) \text{ for all } i, j, k, m.$$

With this assumption, we need to determine only $|W| \cdot |V| = 200,000 \cdot 2 = 4 \times 10^5$ which may be feasible given the 2×10^8 words in our training data.

m-estimate of probability

In general, we estimate the conditional probability

$$P(A|B) \text{ by } \frac{n_{A \wedge B}}{n_B}, \text{ where } n_{A \wedge B} \text{ and } n_B \text{ are the number of times } A \wedge B$$

and B occurred in the training data, respectively. Problems arise when $n_{A \wedge B} = 0$. We deal with this problem by assuming

- a non-zero prior estimate p for $P(A|B)$, and
- a number m which is how confident we are in the prior p as

measured in number of samples. m is referred to as the **equivalent sample size**.

Instead of estimating $P(A|B)$ by $\frac{n_{A \wedge B}}{n_B}$, we estimate it by $\frac{n_{A \wedge B} + mp}{n_B + m}$.

Calculating the conditional probabilities

We estimate $P(w_k | v_j)$ by

$$P(w_k | v_j) = \frac{n_k + 1}{n + |Vocabulary|}, \text{ where}$$

- n is the total number of word positions in all training samples whose target value is v_j
- n_k is the number of times word w_k is found among these n word positions, and
- $|Vocabulary|$ is the total number of distinct words found in the training data.

Training a naive Bayes classifier

1) Collect all words that occur in *Examples*

$\mathcal{V}$ *Vocabulary* $\leftarrow$ the set of all distinct words occurring in any text documents from *Examples*.

RISE

Training a naive Bayes classifier ...

2) Calculate the needed $P(v_j)$ and $P(w_k | v_j)$ probability terms

- For each target value $v_j \in V$ do
 - $docs_j \leftarrow$ the subset of documents from *Examples* for which the target value is v_j
 - $P(v_j) \leftarrow \frac{|docs_j|}{|Examples|}$
 - $Text_j \leftarrow$ a single document created by concatenating all the members of $docs_j$
 - $n \leftarrow$ total number of word positions in $Text_j$

Classifying using a naive Bayes classifier

- $positions \leftarrow$ all word positions in Doc that contain words $\in Vocabulary$
- Return v_{NB} , where

$$v_{NB} = \underset{v_j \in V}{\operatorname{argmax}} P(v_j) \prod_{i \in Positions} P(w_i | v_j)$$

Moby Dick fun facts

All comparisons are to the books contained in Project Gutenberg, which has 56,920 free eBooks.

- 211,763 words
- Vocabulary of 17,227 words (12th largest)
- Vocabulary density is 0.081 (9th most dense of works over 30,000 words)
- Most frequent words (compared to averages)
 - whale, sperm, harpooner, Pequod, Leviathan, fishery
- Least frequent words (compared to averages)
 - miss, fortune, happiness, smiled, angry, enemies

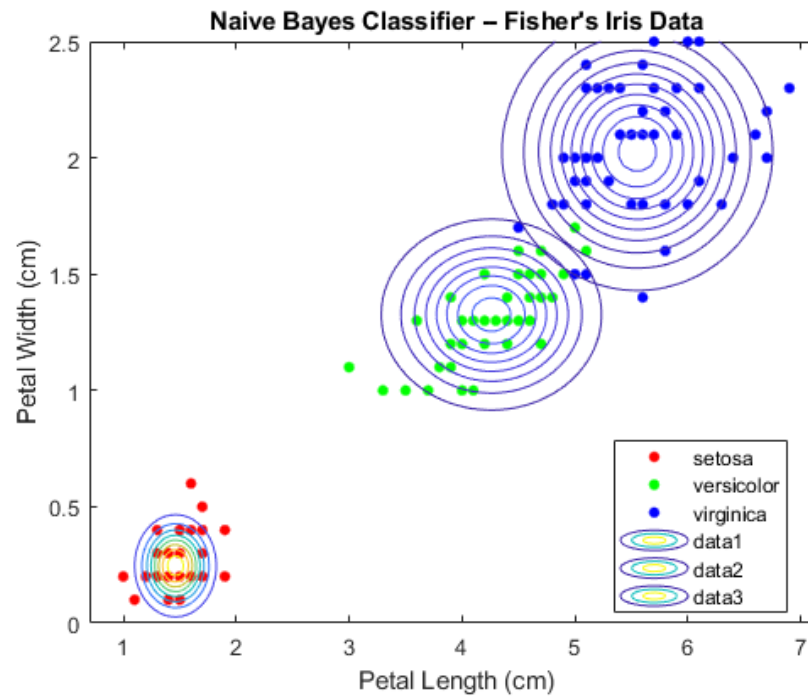
from www.mine-control.com/zack/guttenberg

Ending of *Moby Dick*

Buoyed up by that coffin, for almost one whole day and night, I floated on a soft and dirgelike main. The unharmed sharks, they glided by as if with padlocks on their mouths; the savage sea-hawks sailed with sheathed beaks. On the second day, a sail drew near, nearer, and picked me up at last. It was the devious-cruising Rachel, that in her retracing search after her missing children, only found another orphan.

Moby Dick or, The Whale
by Herman Melville
1851

Another application of a Naïve Bayes classifier



<https://www.mathworks.com/help/stats/fitcnb.html>

Conclusions

- Bayes Theorem links the conditional probabilities of two related events.
- Using Bayes Theorem we can update the probability of an outcome from what we know at one point (the prior probability) to what we know after new observations have been made (the posterior probability). Examples of new observations include the results of a medical test and training data,
- A number of different machine learning algorithms are based on the application of Bayes Theorem. Among these algorithms are
 - Maximum a posteriori hypothesis learning in which the most probable hypothesis is used.
 - Maximum likelihood learning in which the hypothesis that maximizes the likelihood of the data given a hypothesis is used.
 - Bayes optimal learning in which the most probable outcome is used.
 - Bayes optimal learning using the Gibbs algorithm in which a hypothesis is chosen at random according to the probabilities of the hypotheses.
 - Naïve Bayes learning in which it is assumed that attribute values are conditionally independent of the target value.